

Honors Intro Micro | Homework 5

Due Sunday Nov. 14

Homework is designed to both test your knowledge and challenge you to apply familiar concepts in new applications. Answer clearly and completely; show your work so I can understand your thought process for partial credit; you are welcomed and encouraged to work in groups as long as your work is your own.

Note

If you submitted Question 2 on Homework 4 you don't need to answer Question 1 here.

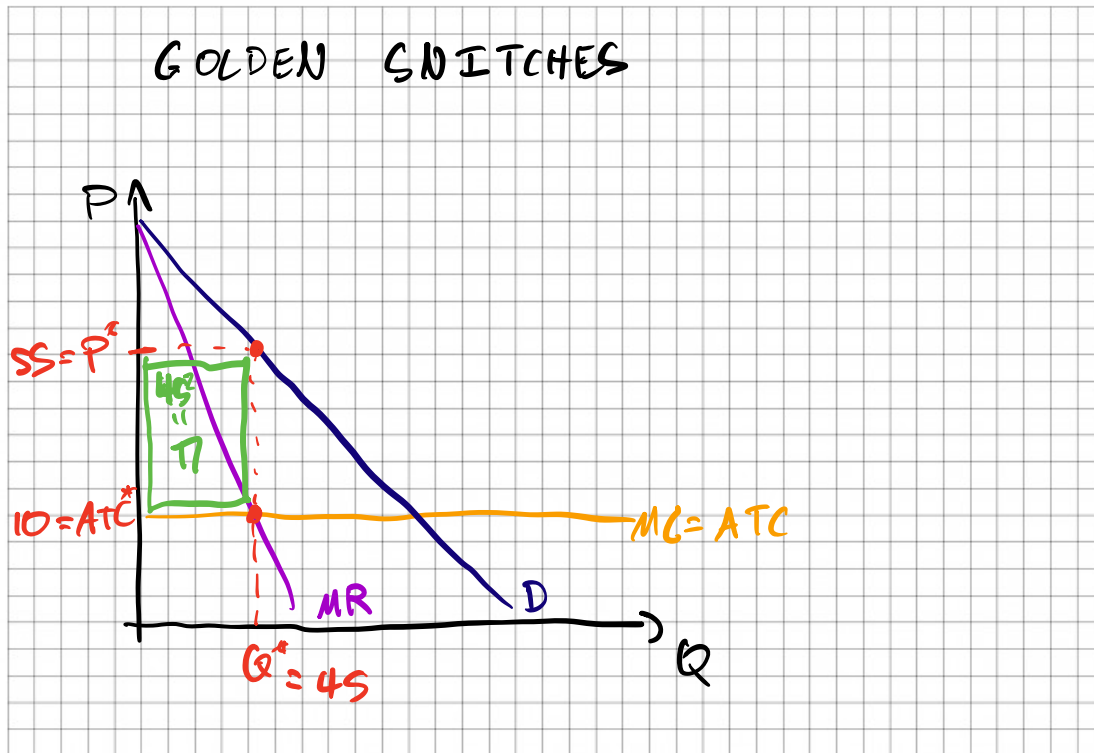
Question 1 (HW4,Q2) | Golden Snitches

The golden Snitch is another piece of equipment in the game of Quidditch. Since manufacturing the Snitch to the uniformity required for competition is incredibly difficult, everyone in the sporting world buys their golden Snitches from one seller: Ruffelbottom's Quidditch Supply. The demand for golden Snitches is given by:

$$P = 100 - Q$$

where prices are given in terms of galleons. Ruffelbottom's marginal cost of producing each Snitch is a constant 10 galleons, and there are no fixed costs. The marginal revenue per Snitch is:

$$MR = 100 - 2Q$$



Part A. Quantity

What is the profit maximizing number of Snitches Ruffelbottom should sell? Label on the graph above.

$$MC = MR$$

$$10 = 100 - 2Q$$

$$2Q = 90$$

$$Q^* = 45$$

Part B. Price

What price should Ruffelbottom charge per Snitch? Label on the graph above.

$$P = 100 - Q = 100 - 45$$

$$P^* = 55$$

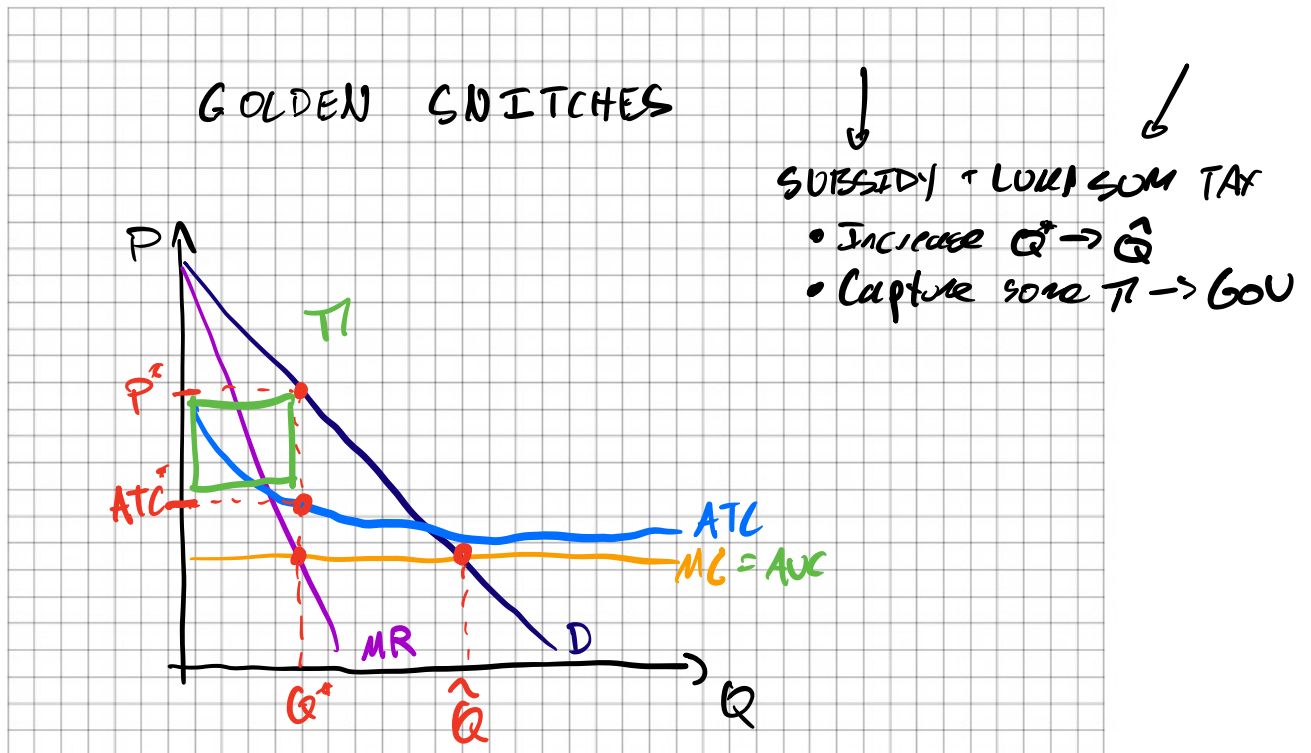
Part C. Profit

What is Ruffelbottom's profit from selling Snitches? Label on the graph above.

$$\pi = 45(55 - 10) = 45^2 = \pi$$

Part D. Lumpsum Tax

The Ministry realized Ruffelbottom was capturing a great deal of welfare from the quidditch world, and decided to impose a lumpsum tax on the seller. Use a graph to show the affect this lumpsum tax had on the market.



Question 2 | Broomsticks

Broomsticks are a branded product in the quidditch community, with each of the many broomstick makers producing broomsticks with slightly different handling and acceleration characteristics. Cleansweep Broom Company had been in the business for a long time and few, if any, new broomstick makers had opened. This all changed when the Ministry imposed a fixed annual subsidy of 100 silver pennies on each broomstick maker. Using a graph to illustrate your answer, tell the story of the broomstick market and the impact of this fixed annual ~~subsidy~~ subsidy. Be sure to finish in a long run equilibrium.

